

NEB Class 12 Management Math

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Unit 1: Basic Principles of Counting, Permutations, and Combinations

Basic Principles of Counting

1. The Multiplication Principle (Fundamental Principle of Counting)

- If one event can occur in m ways, and a second in n ways, the total number of ways for both to occur is $m \times n$.
- **Example:** A restaurant offers 3 main courses and 4 desserts. The total number of meal combinations is $3 \times 4 = 12$.
- **Example 2:** How many two-digit numbers can be formed using the digits 1, 2, 3, 4, and 5 if repetition is allowed? The number of ways is $5 \times 5 = 25$.

2. The Addition Principle

- If two events are mutually exclusive and can occur in m and n ways respectively, the total number of ways for *either* event to occur is $m + n$.
- **Example:** A student can choose a book from 5 novels or 3 comic books. The total number of choices is $5 + 3 = 8$.
- **Example 2:** A student can choose a course from either the 4 science electives or the 6 humanities electives. The total number of choices is $4 + 6 = 10$.

Permutations

- **Definition:** An arrangement of objects in a specific order.
- **Key Idea:** Order matters.
- **Notation:** $P(n, r)$ or ${}_nP_r$
- **Formulas:**

– Permutation of n distinct objects taken r at a time:

$$P(n, r) = \frac{n!}{(n-r)!}$$

– Permutation of n distinct objects taken all at a time:

$$P(n, n) = n!$$

- **Example:** How many ways to arrange 3 people in 3 chairs? $P(3, 3) = 3! = 6$ ways.
- **Example 2:** In a race with 10 runners, how many ways can the first, second, and third place be awarded? $P(10, 3) = \frac{10!}{(10-3)!} = \frac{10!}{7!} = 10 \times 9 \times 8 = 720$ ways.

Permutations with Repetition & Circular Permutations

1. Permutations with Repetition

- For n objects with p_1 of one kind, p_2 of a second, etc., the number of distinct permutations is:

$$\frac{n!}{p_1!p_2!\dots}$$

- Example:** Arrangements of "APPLE": $\frac{5!}{2!} = 60$.
- Example 2:** How many distinct arrangements of the letters in the word "MISSISSIPPI"? The word has 11 letters, with M (1), I (4), S (4), P (2). The number of arrangements is $\frac{11!}{1!4!4!2!} = 34,650$.

2. Circular Permutations

- Number of ways to arrange n distinct objects in a circle is $(n-1)!$.
- Example:** Seating 5 people around a table: $(5-1)! = 4! = 24$.
- Example 2:** How many ways can 6 different colored beads be arranged on a necklace? Since a necklace can be flipped, the number of arrangements is $\frac{(6-1)!}{2} = \frac{5!}{2} = \frac{120}{2} = 60$.

Combinations

- Definition:** A selection of objects where the order does not matter.
- Key Idea:** Order does not matter.
- Notation:** $C(n, r)$ or ${}_nC_r$ or $\binom{n}{r}$
- Formula:** Combination of n distinct objects taken r at a time:

$$C(n, r) = \frac{n!}{r!(n-r)!}$$

- Example:** How many ways to choose a committee of 3 from 5 people?

$$C(5, 3) = \frac{5!}{3!2!} = 10 \text{ ways.}$$

- Example 2:** A student must answer 5 questions out of 10 on an exam. How many different choices of questions does the student have? $C(10, 5) = \frac{10!}{5!(10-5)!} = \frac{10!}{5!5!} = 252$ ways.

Permutations vs. Combinations - Summary

Feature	Permutation	Combination
Order	Matters	Does not matter
Question Type	Arrangements, rankings	Selections, committees
Formula	$P(n, r) = \frac{n!}{(n-r)!}$	$C(n, r) = \frac{n!}{r!(n-r)!}$

Practice Questions from Board Exams

- A student has to answer 8 questions out of 10. How many choices does he have if he must answer at least 4 questions from the first 5 questions?
- In how many ways can the letters of the word 'MANAGEMENT' be arranged?
- Find the value of n if $P(n, 4) = 12 \times P(n, 2)$.
- A committee of 4 is to be formed from 5 gentlemen and 6 ladies. In how many ways can this be done if the committee contains (a) at least 1 lady and (b) at most 2 ladies?
- How many different numbers can be formed with the digits 1, 2, 3, 4, 5, 6, and 7, so that no two odd digits come together?

6. Six persons are to be seated around a circular table. In how many ways can this be done if two particular persons must sit together?
7. A box contains 5 red balls and 3 white balls. In how many ways can we select 3 balls such that at least 2 are red?
8. Prove that $C(n, r) + C(n, r - 1) = C(n + 1, r)$.
9. In how many ways can the letters of the word 'MATHEMATICS' be arranged? How many of these arrangements begin with 'C' and end with 'S'?
10. If $P(n, 3) = 10 \times C(n, 4)$, find the value of n .